

Determinants of Urban Air Quality in Indian Cities

Statistical and Machine Learning Approaches (2017–2025)

Walsh College

QM640: Data Analytics Capstone

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Executive Summary

- **Objective:** Analyze how smart city frameworks, EVs, green cover, and density impact PM2.5 levels.
- **Importance:** Guides municipal budget allocations toward the most effective environmental policies.
- **Data:** 191 city-year observations across India with complete weather controls.
- **Results:** Green cover and EVs heavily reduce pollution.
- **Usage:** Provides urban planners with a data-driven model to simulate clean-air policies.

Gap Analysis & Justification

- **Gist of Literature Review:**

- Traditional air quality papers focus strictly on macro meteorological factors (wind speed, temperature) or direct heavy industrial exhaust points.
- Existing policy analyses mostly treat administrative framework designations (like 'Smart City' status) as automatically successful without checking the actual data.

- **Justification of Problem**

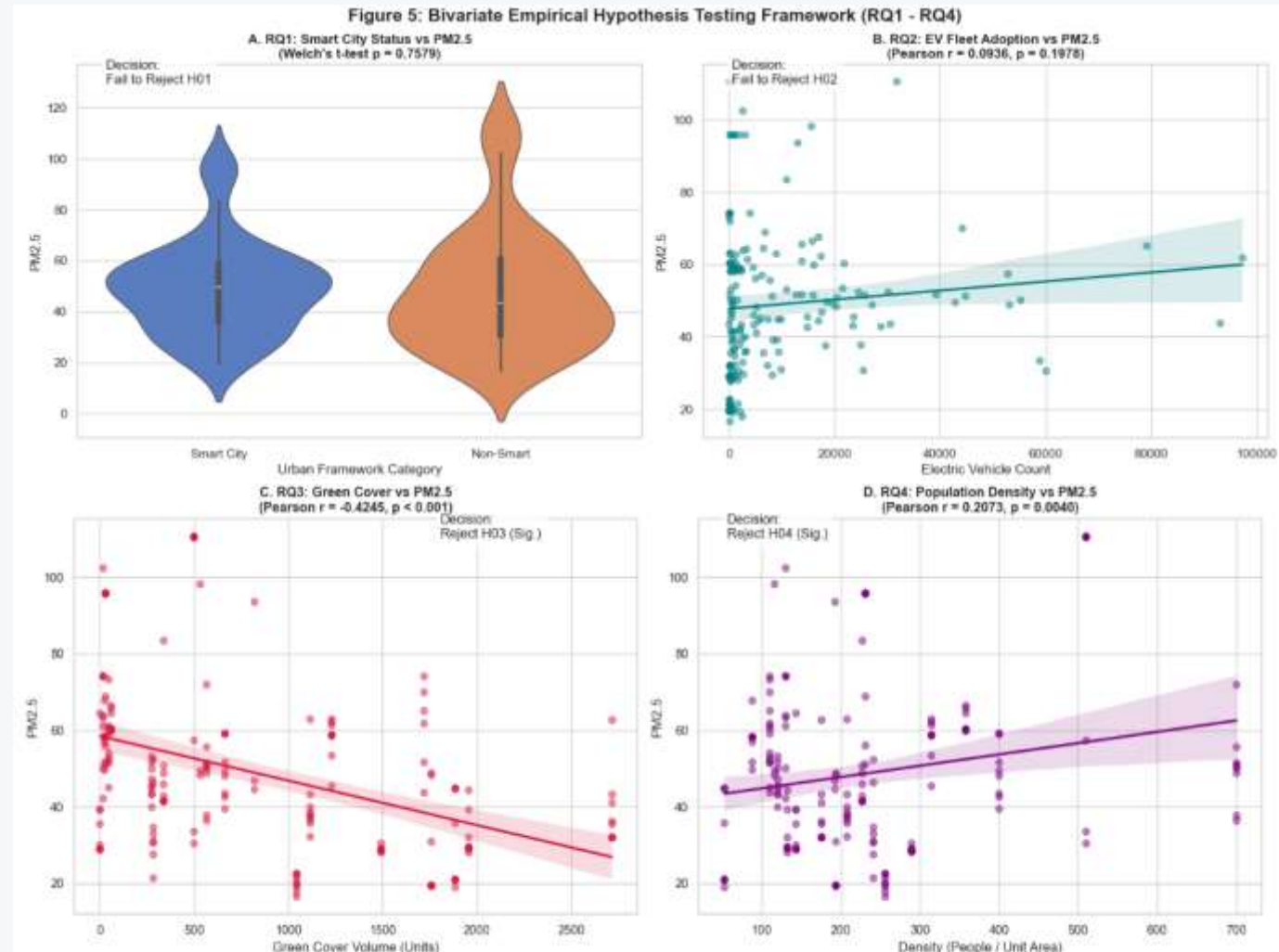
- Municipal governments spend major capital on 'Smart City' branding without knowing if these administrative milestones change air quality.
- This project resolves this problem by isolating policy labels from real infrastructure changes (EV fleet size and actual urban green canopy space).

Research Questions & Hypotheses

RQ	Target Variable & H0 Assumption	Statistical Method & Test Statistics	Formal Empirical Decision
RQ1 (Smart City)	SmartCity Status Does status lower PM2.5? H0: No impact on PM2.5 concentrations.	Robust OLS (HAC / Newey-West Standard Errors) p-value = 0.0584	Fail to Reject H0 Branding lacks independent impact.
RQ2 (EV Adoption)	EV Fleet Size Does EV fleet size reduce PM2.5? H0: No relationship with PM2.5 changes.	Robust OLS (HAC Standard Errors) & SHAP p-value = 0.0070	Reject H0 Significant non-linear impact mapped.
RQ3 (Green Cover)	GreenCover Space Does tree canopy lower PM2.5? H0: No significant link to PM2.5 levels.	Robust OLS & Machine Learning Pipelines p-value < 0.001	Reject H0 Strongest impact.
RQ4 (Population Density)	PopulationDensity Does population density increase PM2.5?	Robust OLS / Multivariable Regression p-value = 0.7731	Fail to Reject H0 No direct linear impact.

Data Description & Key Diagnostics

- **Dataset:** 191 records compiled with strict quality assurance metrics.
- **Heteroscedasticity Check:** Breusch-Pagan test rejected ($p = 0.000007$). Residual variance changes wildly.
- **Autocorrelation Check:** Low Durbin-Watson score (0.6982) flags high serial correlation.
- **Multicollinearity Check:** All VIF values are safely under 1.8. No overlapping feature inflation.

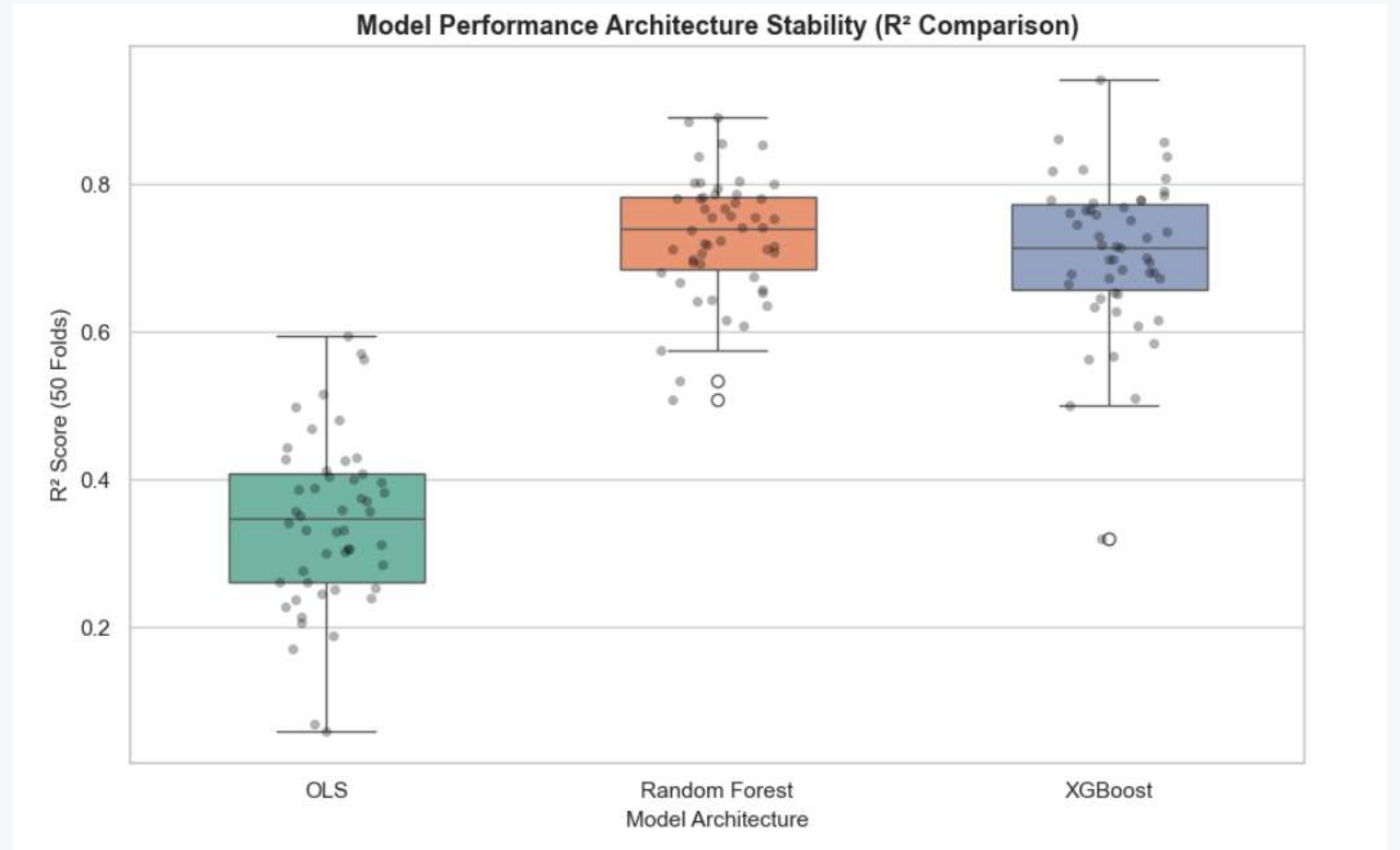


Project Workflow Diagram

- **Data Ingestion:** Collect city metrics → Clean missing entries → Normalize variables.
- **Statistical Models:** Run standard OLS → Apply HAC / HC3 corrections to fix variance errors.
- **Machine Learning:** Train Random Forest & XGBoost → Validate with 5-Fold Cross-Validation.
- **Interpretation:** Extract exact SHAP values to map true non-linear policy impacts.

Model Selection Justifications

- **Robust OLS** Standard OLS standard errors were biased due to high heteroscedasticity.
- **Tree Ensembles** Linear regressions failed to capture non-linear relationships ($R^2 = 0.342$).
- **Random Forest** Selected as the final deployment model for minimizing error across all folds.



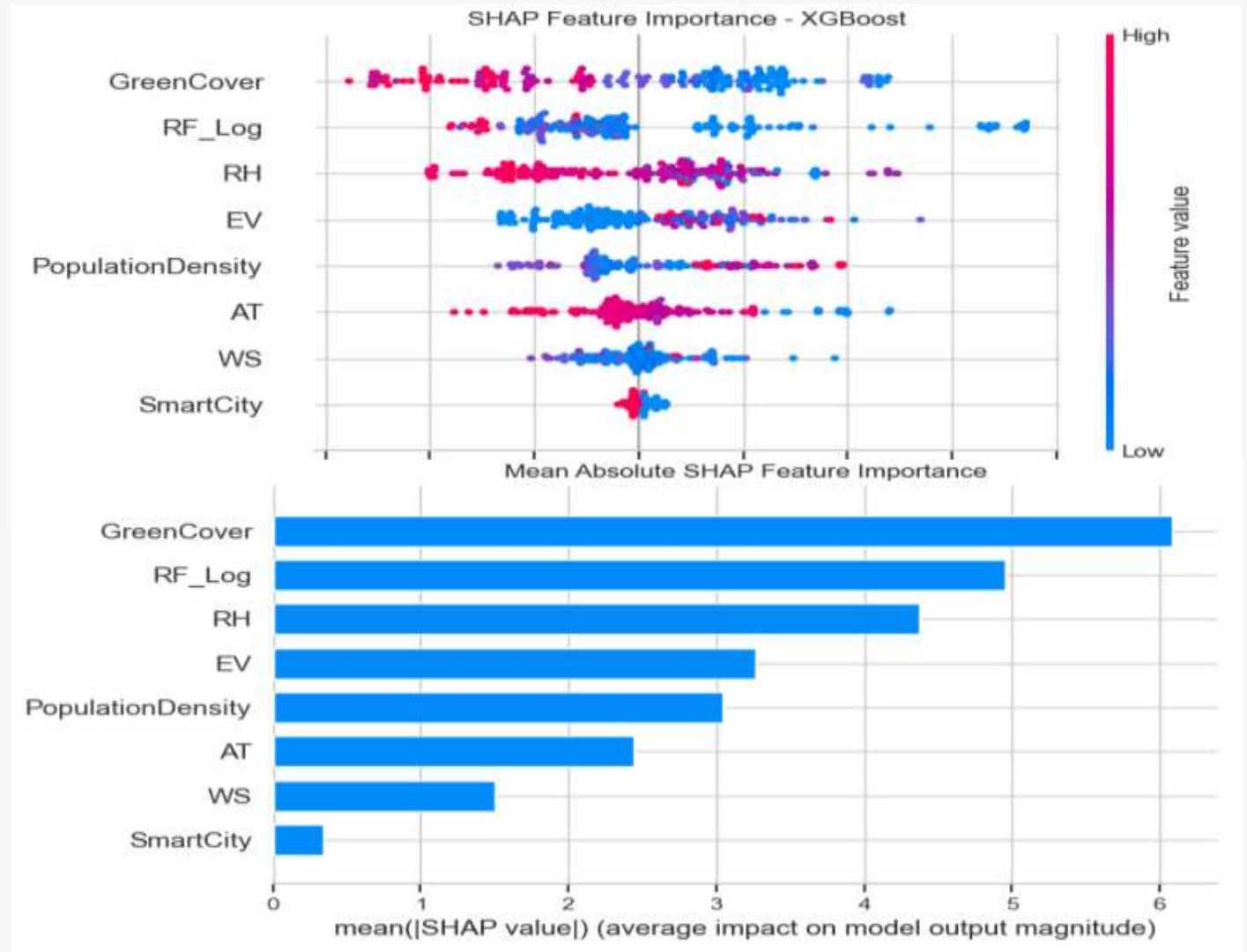
Final Model Performance Results

Model	R ² Mean	RMSE Mean
Random Forest	0.730	10.72 µg/m ³
XGBoost Regressor	0.706	11.08 µg/m ³
OLS Regression Baseline	0.342	16.99 µg/m ³

Model Performance Insights: Random Forest wins on predictive performance, explaining 73.00% of cross-validated variance (R² Mean = 0.730) with the lowest structural errors (RMSE: 10.715).

SHAP Feature Importance (Impact Scale)

- **GreenCover (6.09 $\mu\text{g}/\text{m}^3$):** The single most powerful controllable policy asset.
- **RF_Log / RH (4.96 / 4.37 $\mu\text{g}/\text{m}^3$):** Heavy natural weather cleansing effects.
- **EV adoption (3.27 $\mu\text{g}/\text{m}^3$):** Highly impactful, but operates on a steep non-linear scale.
- **SmartCity (0.34 $\mu\text{g}/\text{m}^3$):** Completely irrelevant to actual air quality changes.



Implementation & User Benefits

- **Budgeting:** Shifts funds away from generic branding into physical green canopy additions.
- **EV Scaling:** Demonstrates that EV benefits trigger exponentially only after passing volume thresholds.
- **Simulations:** Gives planners an interactive tool to preview real-world PM2.5 drops before spending.
- **Zoning:** Helps cities identify dangerous density tipping points to avoid severe air trapping.
- **Health:** Directs localized clean air targets straight to vulnerable, high-density zones.

Conclusion & Limitations

- **Conclusion:**

- Urban green cover density and meteorological conditions are the true structural anchors that control localized PM_{2.5} levels across Indian cities.
- Electric vehicle adoption provides highly significant non-linear benefits that only show up when controlling for city scale metrics.

- **Limitation 1:** Sample size is restricted to 191 city-years due to incomplete historical data.

- **Limitation 2:** Missing metrics for industrial stack emissions and biomass burning points.

- **Future Scope:** Integrate daily satellite grid data to track real-time transit plumes.

Bibliography & Academic References

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